

24/07/2026

HW:-

Q.) What are the various problems which you have observed in society & community. Pick any 5 problems. Find out if they can be solved using Data Structures and Algorithms.

The various problems are Traffic congestion, Online fraud, Food waste, emergency medical response delays, inefficient and inequitable public transit, many disasters to relief, power outage management.

(1) Traffic Congestion:-

Traffic jams are common in cities, especially during peak hours. They waste time, increase fuel consumption, and cause air pollution.

Solution: Graph data structures and shortest path

algorithms such as Dijkstra's algorithm can find the fastest and least congested routes. Navigation apps use these algorithms to guide drivers.

(2) Hospital Appointment Management :-

Patients often have to wait for long hours because appointments are not managed efficiently.

Solution :-

Queue data structures and scheduling algorithms help organize patient appointments, reduce waiting times, and improve hospital services.

(3) Online Fraud Detection :-

Online banking and shopping are vulnerable to fraudulent transactions and cybercrime.

Solution :-

Hash Tables, searching algorithms, and pattern-matching techniques quickly analyse transactions and identify suspicious activities.

(4) E-commerce Product Search :-

Customers may struggle to find the products they need from millions of available items.

Solution :-

Trees, hashing, and sorting algorithms provide fast and accurate product searches and recommendations.

(5) Job Portal Matching :-

Job seekers often struggle to find vacancies that match their qualifications and skills.

Solution :-

Matching, searching, and sorting algorithms recommend suitable jobs based on user profiles.